

Chicago_Crimes_2015

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In this R Markdown file, I'll be going over my methodology of producing the visualizations and results presented in my report.

First, I'll start by loading each library I may need.

```
library(plotly)
library(dplyr)
library(tidyverse)
library(gridExtra)
library(ggrepel)
library(GGally)
library(ggmap)
library(readxl)
```

Next, I downloaded and read-in the 2015 crimes data from this website:

<https://data.cityofchicago.org/Public-Safety/Crimes-2015/vwwp-7yr9>

```
Crimes_2015 <- read.csv("C:/Users/[REDACTED]/OneDrive/Desktop/DePaul/Job
Assessments/Civis Analytics/Crimes_-_2015.csv")
summary(Crimes_2015)
```

```
##           ID           Case.Number           Date           Block
##  Min.      : 21714      Length:264613      Length:264613      Length:264613
## 1st Qu.:10033741      Class :character      Class :character      Class :character
## Median :10148588      Mode  :character      Mode  :character      Mode  :character
## Mean    :10136841
## 3rd Qu.:10262962
## Max.    :12585297
##
##           IUCR           Primary.Type           Description
## Location.Description
## Length:264613      Length:264613      Length:264613      Length:264613
## Class :character      Class :character      Class :character      Class :character
## Mode  :character      Mode  :character      Mode  :character      Mode  :character
##
##
##           Arrest           Domestic           Beat           District
## Length:264613      Length:264613      Min.   : 111      Min.   : 1.00
## Class :character      Class :character      1st Qu.: 612      1st Qu.: 6.00
## Mode  :character      Mode  :character      Median :1023      Median :10.00
```

```
##                               Mean   :1144   Mean   :11.21
##                               3rd Qu.:1654   3rd Qu.:16.00
##                               Max.    :2535   Max.    :31.00
##
##      Ward      Community.Area  FBI.Code      X.Coordinate
## Min.   : 1.00    Min.   : 1.00  Length:264613  Min.   :1094231
## 1st Qu.:10.00    1st Qu.:23.00  Class :character 1st Qu.:1152410
## Median :23.00    Median :32.00  Mode  :character Median :1166064
## Mean   :22.81    Mean   :37.58      Mean   :1164457
## 3rd Qu.:34.00    3rd Qu.:57.00      3rd Qu.:1176389
## Max.   :50.00    Max.   :77.00      Max.   :1205111
## NA's    :2      NA's    :6702
## Y.Coordinate      Year      Updated.On      Latitude
## Min.   :1813897    Min.   :2015  Length:264613  Min.   :41.65
## 1st Qu.:1858595    1st Qu.:2015  Class :character 1st Qu.:41.77
## Median :1891472    Median :2015  Mode  :character Median :41.86
## Mean   :1885560    Mean   :2015      Mean   :41.84
## 3rd Qu.:1908452    3rd Qu.:2015      3rd Qu.:41.91
## Max.   :1951523    Max.   :2015      Max.   :42.02
## NA's    :6702      NA's    :6702
## Longitude      Location
## Min.   : -87.93  Length:264613
## 1st Qu.: -87.72  Class :character
## Median : -87.67  Mode  :character
## Mean   : -87.67
## 3rd Qu.: -87.63
## Max.   : -87.53
## NA's    :6702
```

I'll now start pre-processing this data. First, I'll save the file in a new dataframe called `chicago2015_df`. Next, I renamed the primary.type to offense for clarity, removed unneeded columns such as ID, case number, description, year, and the UTM coordinates. The ID and case number are unique and not needed in such an analysis. The descriptions aren't needed either since I'll be focusing more on the offense itself. We do not need the year column since all the data provided is of the year 2015. The UTM X and Y coordinates aren't needed since we're already provided with the latitude and longitude of each crime.

```
#saving the file in a new dataframe called chicago2015_df
chicago2015_df <- Crimes_2015

#rename primary type to offense for clarity
chicago2015_df <- chicago2015_df %>% rename(offense = Primary.Type)

#removed ID, case number, description, year, X and Y coordinates
chicago2015_df <- chicago2015_df %>% select(!c(ID, Case.Number, Description,
Year, X.Coordinate, Y.Coordinate))

#converted blank values to NA
chicago2015_df <- chicago2015_df %>% mutate_all(list(~na_if(., "")))
```

We can clean it up a bit more by removing any data with missing values or NA. This dataset is quite large so, as long as we're not compromising the quality of the dataset, we should be good to go!

```
#check for any missing values
chicago2015_df %>% summarise_all(~sum(is.na(.)))

##   Date Block IUCR offense Location.Description Arrest Domestic Beat
## District
## 1      0      0      0          0          573      0      0      0
## 0
##   Ward Community.Area FBI.Code Updated.On Latitude Longitude Location
## 1      2              0          0          0      6702      6702      6702

#total number of entries with missing values
rows_before <- nrow(chicago2015_df)
sprintf("Total Rows: %d", rows_before)

## [1] "Total Rows: 264613"
```

We have missing fields in location.description, wards, longitude, and latitude, and we have 264613 crimes total. Thus, we can safely remove rows with missing data without compromising the quality of our dataset.

```
#removed all NA values
chicago2015_df <- chicago2015_df[complete.cases(chicago2015_df),]

#confirm that they're actually removed
chicago2015_df %>% summarise_all(~sum(is.na(.)))

##   Date Block IUCR offense Location.Description Arrest Domestic Beat
## District
## 1      0      0      0          0          0      0      0      0
## 0
##   Ward Community.Area FBI.Code Updated.On Latitude Longitude Location
## 1      0              0          0          0      0      0      0      0

rows_before <- nrow(chicago2015_df)
sprintf("Total Rows: %d", rows_before)

## [1] "Total Rows: 257768"
```

We're now down to 257768 crimes total and have successfully remove all rows with missing data.

As for the last step, I'll create a separate column for the months to help calculate the frequency of crimes per month.

```
#getting the first two numbers from the Date column and inputting them into a new column
chicago2015_df$months <- substr(chicago2015_df$Date, 1, 2)
head(chicago2015_df)
```

```
##                               Date                Block IUCR  offense
## 1 09/05/2015 01:30:00 PM          043XX S WOOD ST 0486  BATTERY
## 2 09/04/2015 11:30:00 AM    008XX N CENTRAL AVE 0870   THEFT
## 3 09/05/2015 12:45:00 PM     035XX W BARRY AVE 2023 NARCOTICS
## 4 09/05/2015 01:00:00 PM    0000X N LARAMIE AVE 0560   ASSAULT
## 5 09/05/2015 10:55:00 AM    082XX S LOOMIS BLVD 0610  BURGLARY
## 6 09/04/2015 06:00:00 PM    021XX W CHURCHILL ST 0620  BURGLARY
##   Location.Description Arrest Domestic Beat District Ward Community.Area
## 1      RESIDENCE      false      true   924         9    12          61
## 2      CTA BUS      false     false  1511        15    29          25
## 3      SIDEWALK      true      false  1412        14    35          21
## 4      APARTMENT     false      true  1522        15    28          25
## 5      RESIDENCE     false     false   614         6    21          71
## 6  RESIDENCE-GARAGE  false     false  1434        14    32          24
##   FBI.Code      Updated.On Latitude Longitude
## 1    08B 02/10/2018 03:50:01 PM 41.81512 -87.67000
## 2     06 02/10/2018 03:50:01 PM 41.89508 -87.76540
## 3    18 02/10/2018 03:50:01 PM 41.93741 -87.71665
## 4    08A 02/10/2018 03:50:01 PM 41.88190 -87.75512
## 5     05 02/10/2018 03:50:01 PM 41.74438 -87.65843
## 6     05 02/10/2018 03:50:01 PM 41.91464 -87.68163
##                               Location months
## 1 (41.815117282, -87.669999562)      09
## 2 (41.895080471, -87.765400451)      09
## 3 (41.937405765, -87.716649687)      09
## 4 (41.881903443, -87.755121152)      09
## 5 (41.744378879, -87.658430635)      09
## 6 (41.914635603, -87.681630909)      09
```

I think I'm done preprocessing this data and I'm ready to analyze it. First, I'll create a visual to see the 20 most frequent crimes in Chicago by offense.

```
#create a new table for crime frequencies
crime_freq <- as.data.frame(table(chicago2015_df$offense))
crime_freq <- crime_freq %>% filter(Freq>0)

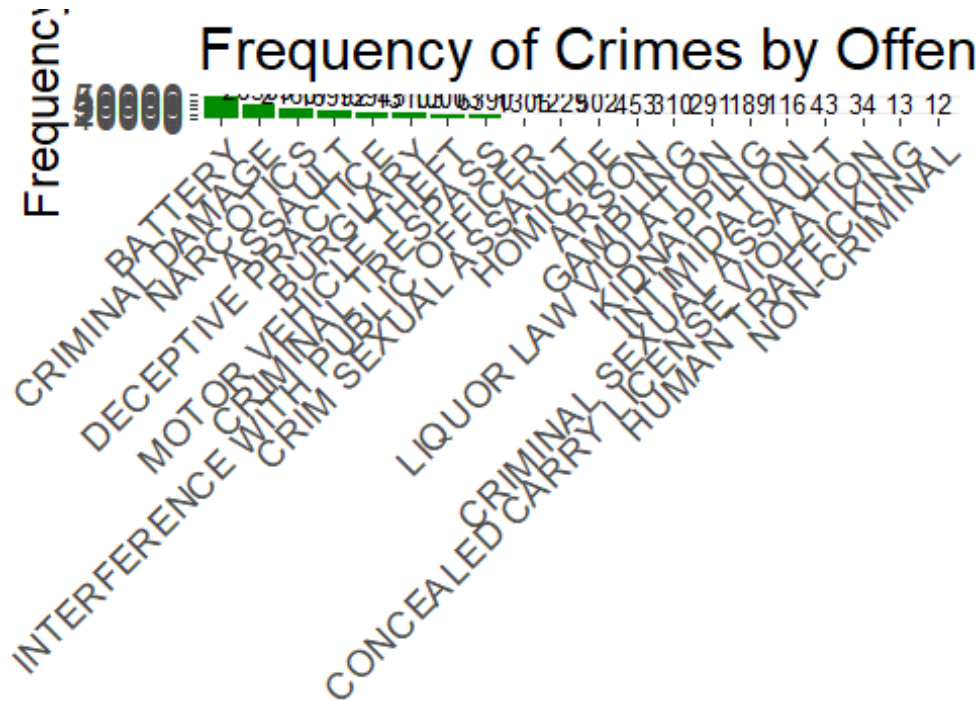
#get the top 20 crimes
crime_freq <- crime_freq[1:20,]

#set plot size
options(repr.plot.width = 14, repr.plot.height = 8)

#create the bar plot
freq_bar <- ggplot(crime_freq, aes(x=reorder(Var1,-Freq), y=Freq)) +
  geom_bar(stat = "identity", fill="green4") +
  labs(title="Frequency of Crimes by Offense (top 20)",
x="", y="Frequency") +
  geom_text(aes(label=Freq), vjust=-0.3, size=3.5)+
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
```

```
text = element_text(size = 18))
```

freq_bar



We can also get these in a numerical format for all offenses:

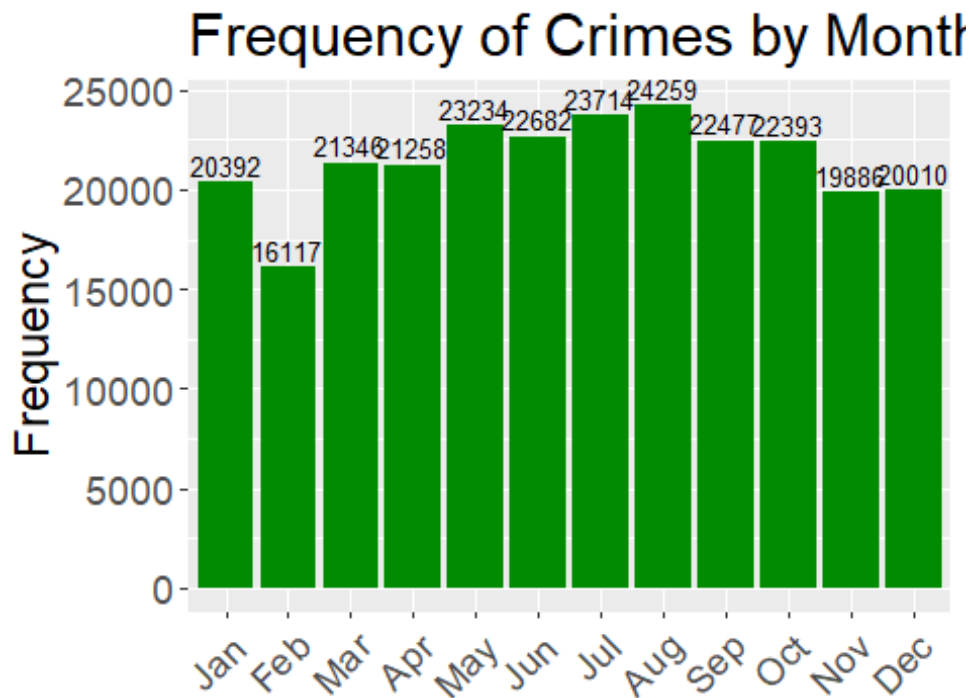
```
chicago2015_df %>% group_by(offense) %>% summarise("count"=n())
```

```
## # A tibble: 33 x 2
##   offense                                count
##   <chr>                                <int>
## 1 ARSON                                453
## 2 ASSAULT                             16992
## 3 BATTERY                             48821
## 4 BURGLARY                            13103
## 5 CONCEALED CARRY LICENSE VIOLATION    34
## 6 CRIM SEXUAL ASSAULT                   1229
## 7 CRIMINAL DAMAGE                       28589
## 8 CRIMINAL SEXUAL ASSAULT                43
## 9 CRIMINAL TRESPASS                     6390
## 10 DECEPTIVE PRACTICE                 13945
## # ... with 23 more rows
```

Let's visualize what the frequency of crimes looks like each month.

```
#create a new table
crime_freq_month <- as.data.frame(table("month" = chicago2015_df$months))
monthnames <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug",
"Sep", "Oct", "Nov", "Dec")
```

```
#plot the frequency per month
month_bar <- ggplot(crime_freq_month, aes(y=Freq, x=month)) +
  geom_bar(stat="identity", fill="green4") +
  scale_x_discrete(label=monthnames) +
  labs(title="Frequency of Crimes by Month", x="", y="Frequency") +
  geom_text(aes(label=Freq), vjust=-0.3, size=3.5)+
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        text = element_text(size = 18), legend.position = "none")
month_bar
```



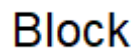
It might be interesting to look into crimes per block.

#Now Let's Look into the frequency of crimes by block.

```
crime_by_block <- as.data.frame(table("Block" = chicago2015_df$Block))
crime_by_block <- crime_by_block %>%
  filter(Freq>0) %>%
  arrange(desc(Freq))
crime_by_block <- crime_by_block[1:20,]

# plot
block_bar <- ggplot(crime_by_block, aes(y=Freq, x=reorder(Block, -Freq))) +
  geom_bar(stat="identity", fill="green4") +
  labs(title="Frequency of Crimes by Block (Top 20)",
       x="Block", y="Frequency") +
  geom_text(aes(label=Freq), vjust=-0.3, size=3.5)+
```

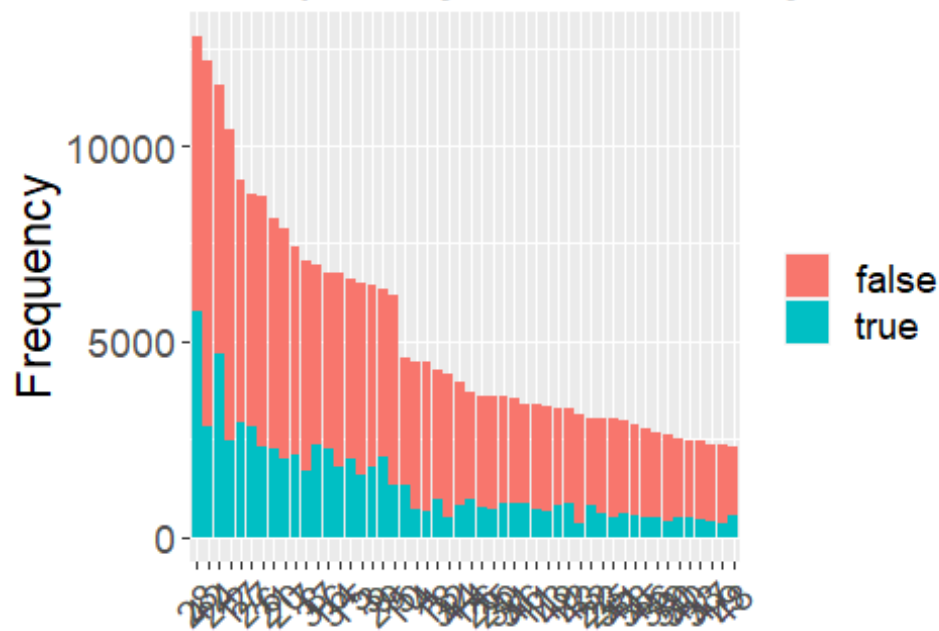
block_bar



```
crime_freq_ward <- as.data.frame(table("ward" = chicago2015_df$Ward, "arrest"
= chicago2015_df$Arrest))
crime_freq_ward <- crime_freq_ward %>% filter(Freq>0)

ward_bar <- ggplot(crime_freq_ward, aes(fill=arrest, y=Freq, x=reorder(ward,
Freq))) +
  geom_bar(position="stack", stat="identity") +
  labs(title="Frequency of Crimes by Ward & Arrests Made",
x="", y="Frequency") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        text = element_text(size = 18),
legend.title=element_blank())
ward_bar
```

Frequency of Crimes by Ward



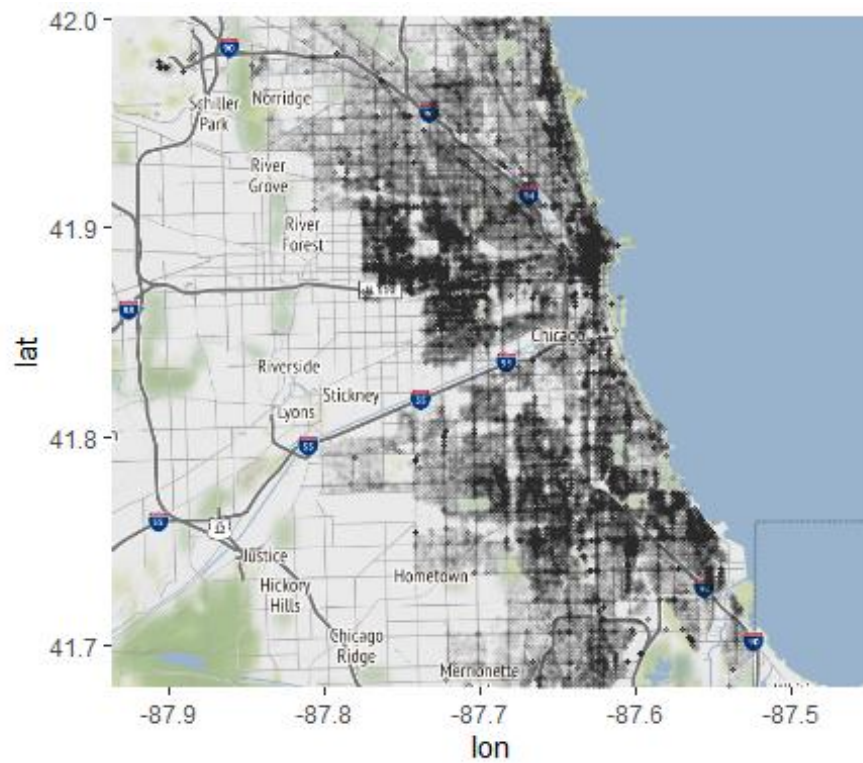
Finally, I'll visualize the density of crimes across Chicago using ggmap.

```
#store bounding box coordinates
chi_bb <- c(
  left = -87.936287,
  bottom = 41.679835,
  right = -87.447052,
  top = 42.000835
)

chicago_map <- get_stamenmap(
  bbox = chi_bb,
  zoom = 11
)

chicago <- chicago_map

ggmap(chicago) +
  geom_point(
    data = chicago2015_df,
    aes(x = Longitude, y = Latitude),
    size = 0.8,
    alpha = .01
  )
```

This sums up what I'm doing in R for now. Next, I'll go into researching a bit more about Chicago to write up the report.